Joydip Dutta

Bolpur, West Bengal, 731204 | +91 8759612725
joydip.dutta9943@gmail.com | GitHub | LinkedIn | LeetCode

Experience

Software Development Engineer Jan. 2026 – Present
Gunpowder Innovations *London, UK (Remote)*

- Architected a robust **Row-Level Security (ACL)** system for a multi-tenant platform across 61 Prisma models, using Redis-backed caching and specific handlers to keep data isolated and secure.
- Integrated **WealthKernel** and **Plaid** APIs while managing a reliable webhook pipeline that handles 31 different event types via **BullMQ** with secure signature verification.
- Implemented **AES-256-GCM** field-level encryption with zero-downtime key rotation to protect sensitive user data, ensuring strict financial compliance and very low query latency.

Software Development Engineer Jan. 2023 – Dec. 2025
The Internet Folks *Bhopal, MP (Remote)*

- Maintained highly reliable backend services for over 30,000 users, using dynamic database partitioning in **MongoDB** to manage millions of records safely.
- Architected a flexible, multi-language backend system using **TypeScript** and **Go**, directing heavy tracking data to **ClickHouse** and core relational data to **PostgreSQL** for maximum performance.
- Developed a hybrid AI system that combines semantic search (**pgvector**) with multi-LLM generation (**OpenAI, Gemini**), allowing enterprise clients to securely plug in their own AI keys.
- Created a multi-tenant "Fund as a Service" platform using **PostgreSQL** and **Drizzle ORM**, designing a strict financial ledger that accurately tracks capital flows and fees.
- Built a digital agriculture platform featuring a real-time negotiation engine, enabling farmers to agree on contracts and automatically generating personalized, day-by-day crop care schedules.

Software Development Intern June 2022 – Aug. 2022
AUM Capital Market Private Limited *Kolkata, WB*

- Developed a React and Node.js solution to automate government bond calculations and payment reminders, which cut down manual work by about 80%.

Projects

Unified Multi-Modal RAG Agent | ***LangChain, Gemini, MongoDB Atlas, Redis***

- Engineered a production-ready AI system using **LangChain** to handle complex questions about PDFs and images, achieving 85% search accuracy for over 1,000 active users.
- Used **MongoDB Atlas Vector Search** and **Redis** caching to make information retrieval fast, resulting in a 45% reduction in API waiting times.

Agnibina Model School Platform | ***Astro v5, React 19, Tailwind v4***

- Architected a high-performance educational portal using **Islands Architecture** to achieve 100/100 Lighthouse scores while ensuring full **CBSE regulatory compliance** through a data-driven disclosure module.

Technical Details

Languages/Frameworks: TypeScript, Go (Fiber), Bun, Python, Node.js, GraphQL, React, Next.js, Astro
Databases/Storage: PostgreSQL, MongoDB, ClickHouse, Redis, DynamoDB, Prisma, Drizzle, AWS S3
Infrastructure/Security: Docker, Kubernetes, BullMQ, AWS (SQS, SES), AES-256-GCM, HMAC, Ngrok
AI & Search: LangChain, Google Gemini, OpenAI, Vector Search (pgvector, Atlas), RAG Architecture

Education

Bachelor of Technology in Computer Science (CGPA: 7.0) Aug. 2018 – May 2022
University of Engineering and Management *Jaipur, RJ*